

# VISHAK LAKSHMAN SANJEEVIKANI MURUGESH

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Backend & AI Engineer with 7+ years of experience designing large-scale distributed systems, AI-powered agent pipelines, and cloud-native platforms. Proven track record building **multi-agent LLM workflows** using LangChain, LangGraph, and RAG architectures with FAISS vector stores; deploying containerized microservices on **Kubernetes (EKS)** and modernizing enterprise platforms with measurable infrastructure and reliability improvements.

## PROFESSIONAL EXPERIENCE

### Software Engineer – HCLTech

Oct 2018 – Present

#### Client: AccidentFund Groups

- Designed and delivered Python-based backend platforms powering billing, payments, simulation pipelines, HPC workflows, and document automation systems used by **1,000,000+ enterprise users**.
- Engineered a **LLM powered MLOps** on AWS Bedrock for **document retrieval co-pilot** where specialized agents handled document ingestion, classification (RAG with FAISS vector store) that surfaces results through UI chatbot improving document retrieval for Claim Handlers and PolicyHolders by **43%**.
- Built automated **multi-agent LLM pipeline** for invoice generation orchestrated using LangChain and LangGraph connected to SmartCOMM via Rest API. A delivery agent manages multichannel distribution to **100,000+ policyholders** with fault-tolerant state handoffs via AWS SQS event-driven messaging with human-in-the-loop approval gates to prevent duplicate delivery and edge-case reviews.
- Designed FastAPI-based microservices behind Nginx supporting billing operations and ACH/digital payment processing, with **GraphQL APIs** for flexible client querying and **WebSocket** streams for real-time payment.
- Executed proof-of-concept migration from on-prem enterprise platform to **AWS microservices architecture**; provisioned infrastructure using **Terraform** and Helm charts for Kubernetes deployments on EKS. Includes multi-zone autoscaling and GitHub-Actions based release workflows.
- Implemented Jenkins CI/CD pipelines, **Redis** caching, autoscaling infrastructure reducing deployment time by **57%** and production defects by **70%**.
- Configured Prometheus/Grafana observability stack for LLM service monitoring: tracked token usage, agent latency, retrieval accuracy, and pipeline error rates across multi-agent workflows (**MLOps** practices).
- Integrated **AWS Secrets Manager** and IAM role-based access control (RBAC) for secrets management across containerized services; enforced TLS encryption and SSO for all internal API communications.
- Collaborated cross-functionally ML research, platform engineering, product teams defining agent evaluation frameworks, A/B testing of prompt strategies and measuring retrieval precision/recall for RAG pipelines.
- Mentored engineers on scalable Python architecture, LangChain agent patterns, API design, and CI/CD automation best practices.

### Programmer Analyst - Cognizant Technologies Solutions

Jun 2015 – Dec 2016

#### Client: AAA Insurance

- Engineered backend modules for an enterprise policy administration platform using **Java, Spring MVC, REST APIs**, and Python automation tools in an Agile environment.
- Implemented automated test frameworks using Selenium and JUnit; configured **Prometheus/Grafana** for system monitoring and regression test coverage tracking.

## EDUCATION

Master of Science in Computer Science - University of North Carolina, Charlotte, NC

May 2018

Bachelor of Technology in Information Technology - Anna University, India

May 2015

## SKILLS

Languages : Python, Java, Go, TypeScript, React.js, [Angular.js](#), Kotlin

Databases : PostgreSQL, DynamoDB, MongoDB, RDS, Oracle, Redis (in-memory), HA replication & backup

AI/LLM : LangChain, LangGraph, Pydantic AI, FAISS, PyTorch, TensorFlow, Github Copilot

Frameworks/Tools : FastAPI, Django, GraphQL, WebSockets, REST APIs, Microservices, Event-driven (SQS, SNS)

CI/CD & MLOps : Jenkins, GitOps, CI/CD pipelines, MLFlow, LLM monitoring, prompt A/B testing, agent evaluation

Monitoring/Security: Prometheus, Grafana, Datadog, RBAC, SSO, TLS, AWS Secrets Manager, IAM

Cloud Infra : AWS(Bedrock,SQS/SNS,Secrets Manager,IAM),GCP,Terraform,Helm, Docker/Kubernetes

## CERTIFICATIONS

- Certified **AWS AI Practitioner** and **AWS Solutions Architect**